

not only development, but also retail stages of their products. This move will help ensure that only those who have legally purchased the IP from a vendor can access the design and features. To ensure complete security, IP companies usually adopt these following measures:

Locking-based security: This is a simple and direct way to restrict access to the product. It protects the communication of IPs with remote devices by applying encryption-based technology, along with obfuscation and remote activation.

What this basically means is that while the IP is being transferred from the IP or SoC designers to the fabrication facility, the data is encrypted. This data is then decrypted during the fabrication process.

Obfuscation is also used for security measures wherein, the high quality original IP is transformed into a low quality one. This alteration conceals the true nature of the IP being transferred and only an authorized user can convert it to its original state.

Authentication security: Locking-based security is not sufficient as it leaves some loopholes. For instance, the SoC company to which the decryption key is given may sell the same to a third-party. However, biometric security and watermarking provides an additional layer of security. These cannot be cracked as easily as lock-based security protocols.

Some IP companies also deploy fingerprinting technology to ensure that only an authorized member can access the product. In this process, if an IP embedded on an SoC needs to connect to a remote drive, the hardware module on the chip authenticates the external device, prior to establishing the connection. This process ensures that only secure devices are accessed.

Trojan detection: While designers work on an IP, or during the creation of an SoC, the technology may get infected with Trojans, which may compromise the security of the device. In such cases, the device becomes vulnerable to Trojan side channel attacks.

Therefore, it is imperative to detect if any Trojans are present on the IP before it is transferred to the SoC developers. These detection systems would need to be super-efficient to perceive even the smallest traces of Trojan activity.

The technique deployed by the company uses measurements of actual delays in the combinational paths. A Trojan is detected if certain paths show increased delay, when compared to others.